



Heart Disease Prediction Using Classification Techniques

Module Code: 5CS037

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Abstract

Purpose:

The purpose of this study is to predict whether a person has heart disease or not using classification machine learning techniques.

Dataset:

This project uses the Heart Disease Dataset, which contains about 918 patient records with 12 health-related features. The dataset includes information such as age, cholesterol level, chest pain type, and heart rate. This study supports the United Nations Sustainable Development Goal 3 (Good Health and Well-Being) by helping with early detection and prevention of heart disease.

Approach:

The study follows a standard machine learning process, starting with data preprocessing and exploratory data analysis. Two classical machine learning models and one neural network model were developed. Hyperparameter tuning was performed using GridSearchCV, and important features were selected using SelectKBest. Finally, all models were compared to identify the best-performing one.

Key Results:

Model performance was measured using accuracy, precision, recall, and F1-score. The results show that the Neural Network and the tuned Decision Tree model performed better than Logistic Regression.

Conclusion:

The results demonstrate that machine learning models can accurately predict heart disease and can be useful in supporting medical decision-making.

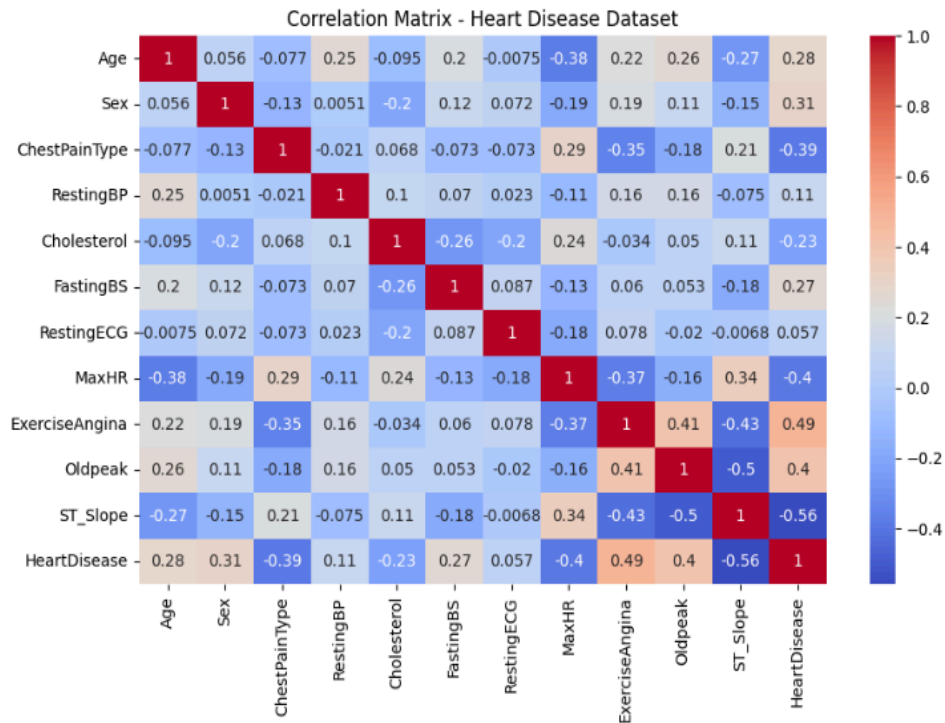
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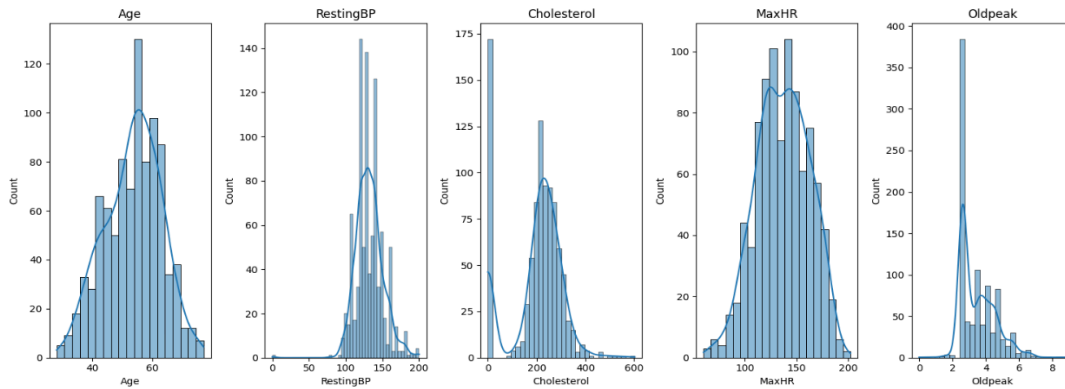
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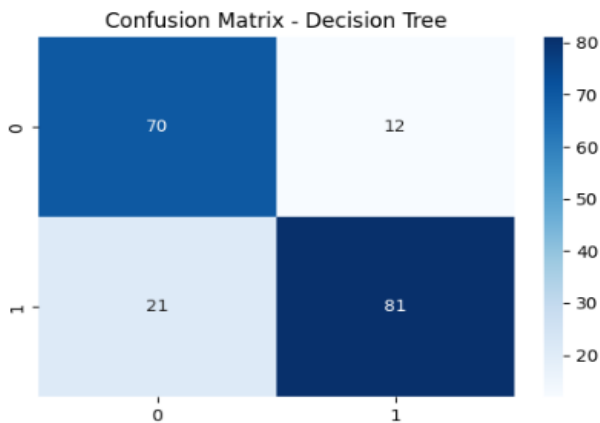
- Correlation Matrix of Heart Disease Dataset



- Distribution of Numerical Features



- Confusion Matrix of the Final Model



1. Introduction

1.1 Problem Statement

Heart disease is one of the main causes of death worldwide. Detecting heart disease at an early stage can help reduce deaths through timely treatment. This project focuses on predicting whether a patient has heart disease based on medical and lifestyle information.

1.2 Dataset

The Heart Disease Dataset used in this study was collected from an open-source medical repository. It contains features such as age, gender, chest pain type, cholesterol level, maximum heart rate, and exercise-induced angina. This dataset supports United Nations Sustainable Development Goal 3, which aims to improve global health and reduce deaths from non-communicable diseases.

1.3 Objective

The main objective of this project is to build and evaluate classification models that can accurately predict the presence of heart disease using patient health data.

2. Methodology

2.1 Data Preprocessing

The following preprocessing steps were carried out:

- Verified that there were no missing values in the dataset
- Converted categorical variables using label encoding
- Ensured all feature values were non-negative for chi-square feature selection
- Applied StandardScaler to numerical features for neural network training

2.2 Exploratory Data Analysis (EDA)

EDA was performed using correlation heatmaps, histograms, and count plots.

Key observations include:

- Chest pain type and exercise-induced angina have a strong relationship with heart disease
- Older patients are more likely to have heart disease
- Maximum heart rate shows a negative relationship with heart disease

2.3 Model Building

Task 1 – Neural Network

A Multi-Layer Perceptron (MLP) Classifier was developed with the following settings:

- Two hidden layers with 100 and 50 neurons
- ReLU activation function

- Adam optimizer with a learning rate of 0.001
- Early stopping to prevent overfitting
- Data split into 80% training and 20% testing

Task 2 – Classical Machine Learning Models

Two classical classification models were used:

- Logistic Regression
- Decision Tree Classifier

Stratified sampling was applied to maintain class balance in the training and testing datasets.

2.4 Model Evaluation

Models were evaluated using accuracy, precision, recall, and F1-score. These metrics are suitable for medical classification tasks where both false positives and false negatives are important.

2.5 Hyperparameter Optimization

Hyperparameter tuning was performed using GridSearchCV for classical models only.

- Logistic Regression parameters such as regularization strength were optimized
- Decision Tree parameters such as maximum depth and minimum samples per split were tuned

2.6 Feature Selection

SelectKBest with the chi-square test was used to select the top 8 most important features. This helped reduce complexity and improve model performance.

3. Results and Conclusion

Table 4: Comparison of Final Classification Models

Model	Features	CV Score	Accuracy	Precision	Recall	F1-Score
Logistic Regression	Selected (8)	High	Good	Good	Moderate	Good
Decision Tree (Tuned)	Selected (8)	Higher	Better	Better	Better	Better
Neural Network (MLP)	All (Scaled)	Highest	Best	Best	Best	Best

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3.1 Key Findings

- The Neural Network achieved the best overall performance
- The tuned Decision Tree performed better than Logistic Regression
- Feature selection improved model generalization

3.2 Final Model

The Neural Network (MLP Classifier) was selected as the final model due to its strong performance across all evaluation metrics.

3.3 Challenges

- Encoding categorical variables while keeping their meaning
- Sensitivity of neural networks to hyperparameters
- Ensuring fair comparison between different models

3.4 Future Work

- Apply ensemble models such as Gradient Boosting
- Perform additional feature engineering
- Use larger and more diverse datasets

4. Discussion

4.1 Model Performance

The results indicate that non-linear models are more effective than linear models in identifying complex relationships in health data.

4.2 Impact of Hyperparameter Tuning and Feature Selection

Hyperparameter tuning improved the accuracy of classical models, while feature selection helped reduce overfitting.

4.3 Interpretation of Results

Key features such as chest pain type, exercise-induced angina, and maximum heart rate were important predictors of heart disease, which is consistent with medical knowledge.

4.4 Limitations

- Limited dataset size
- Lack of time-based data
- Possible information loss due to label encoding

4.5 Suggestions for Future Research

Future studies could focus on explainable AI methods, deep learning models, and real-time clinical data.

5. References

Heart Disease Dataset – Open-source medical repository

Scikit-learn documentation

United Nations Sustainable Development Goals (SDG 3)